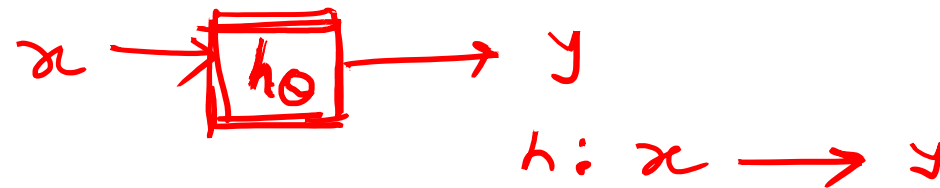


Linear Regression

Linear Regression



- Hypothesis

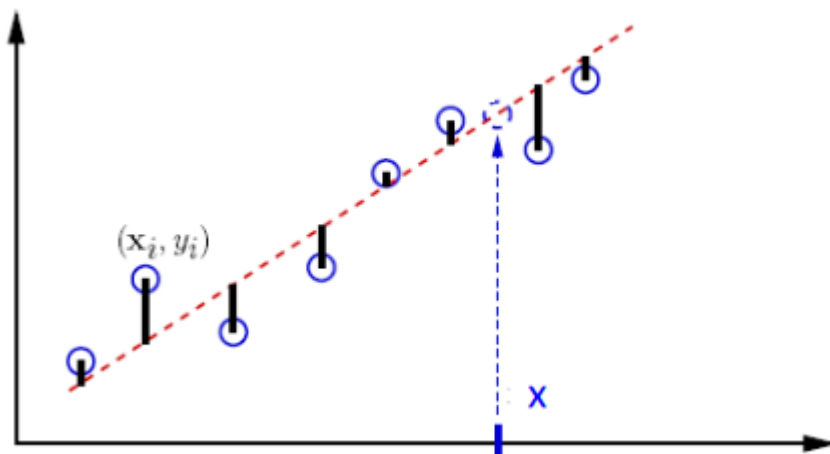
$$\underline{x} = [x_1, x_2, \dots, x_d]^T \rightarrow x = [\overset{1}{x_0}, x_1, \dots, x_d]$$

$$y = \underline{\theta}_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_d x_d = \sum_{j=0}^d \theta_j x_j$$

- Fit a line

$$\theta = [\theta_0, \theta_1, \dots, \theta_d]^T_{d+1}$$

~



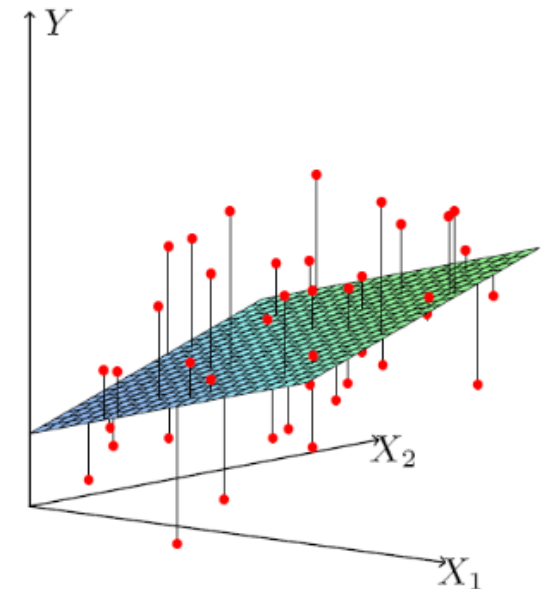
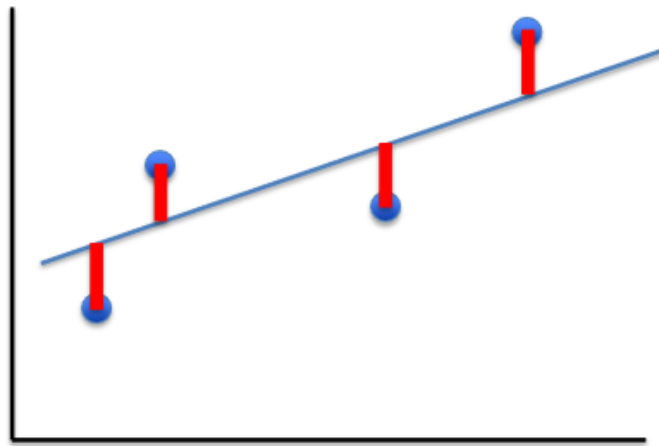
$$\min \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Least Squares Cost Function

- Cost Function

$$J(\boldsymbol{\theta}) = \frac{1}{2n} \sum_{i=1}^n (\underline{h_{\boldsymbol{\theta}}(\mathbf{x}^{(i)})} - y^{(i)})^2$$

Fit by minimizing $J(\boldsymbol{\theta})$

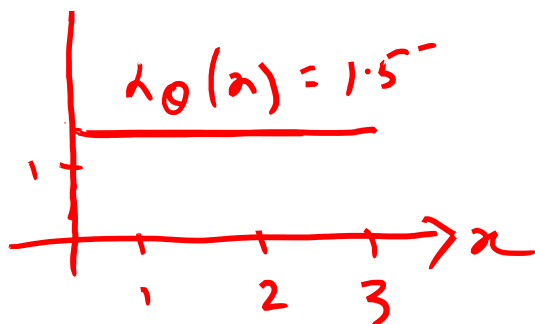


$h_{\theta}(x) = \theta_0 + \theta_1 x$

$\theta_0 = 1.5$

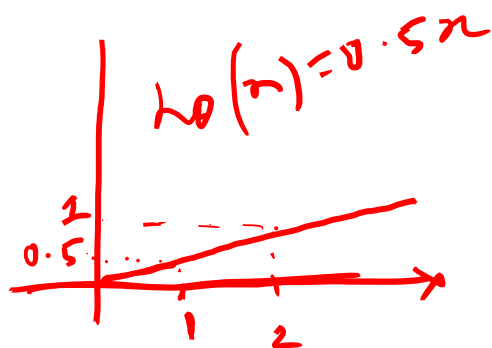
$\theta_1 = 0$

$h_{\theta}(x) = 1.5$



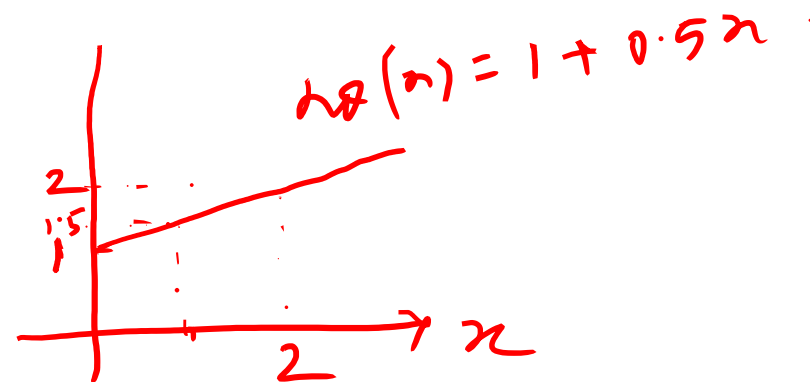
$\theta_0 = 0$

$\theta_1 = 0.5$



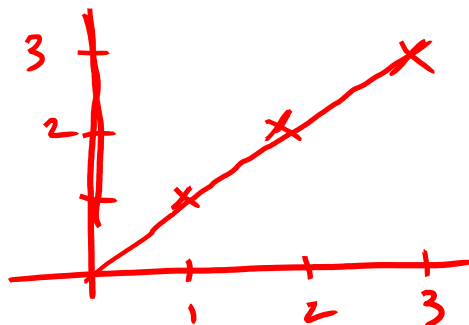
$\theta_0 = 1$

$\theta_1 = 0.5$



x_i

#



$\theta_1 = 1$

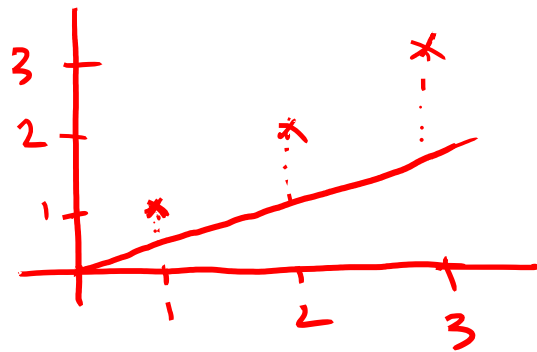
$h_{\theta}(x) = x$

$$J(\theta_1) = \frac{1}{2n} \sum_{i=1}^n (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

$$= \frac{1}{2n} \sum_{i=1}^3 (\theta_1 x^{(i)} - y^{(i)})^2$$

$J(1) = 0 + 0 + 0 = 0$

$$\theta_1 = 0.5$$



$$J(0.5) = \frac{1}{2 \times 3} \left[(0.5 - 1)^2 + (1 - 2)^2 + (1.5 - 3)^2 \right]$$

$$= \frac{1}{6} \left[0.25 + 1 + 2.25 \right]$$

$$= \frac{3.5}{6} = \underline{\underline{0.58}}$$

$$\theta_1 = 1.5$$

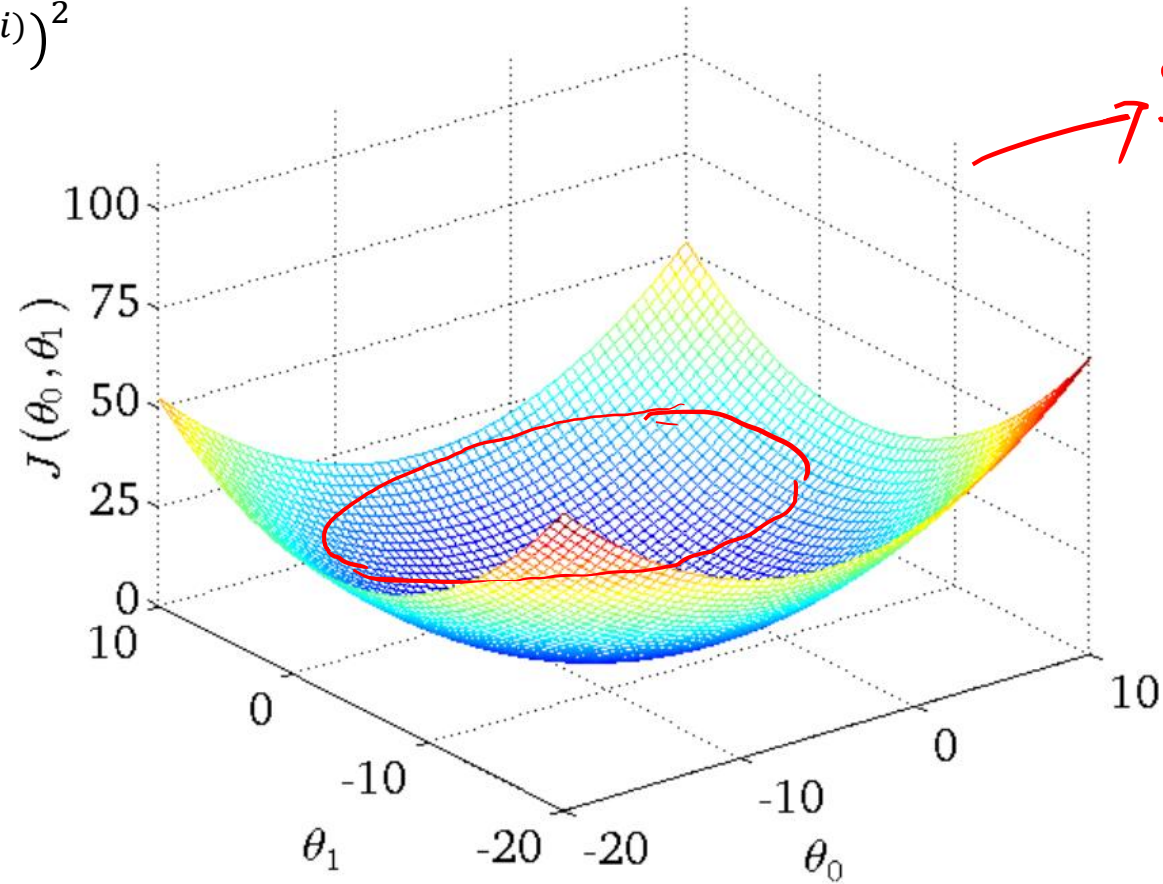
$$J(1.5) = \frac{1}{6} \left[(1.5 - 1)^2 + (3 - 2)^2 + (4.5 - 3)^2 \right]$$

$$= \frac{1}{6} \left[0.25 + 1 + 2.25 \right] = 0.58$$



The shape of the Cost Function

$$J(\boldsymbol{\theta}) = \frac{1}{2n} \sum_{i=1}^n (h_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}) - y^{(i)})^2$$

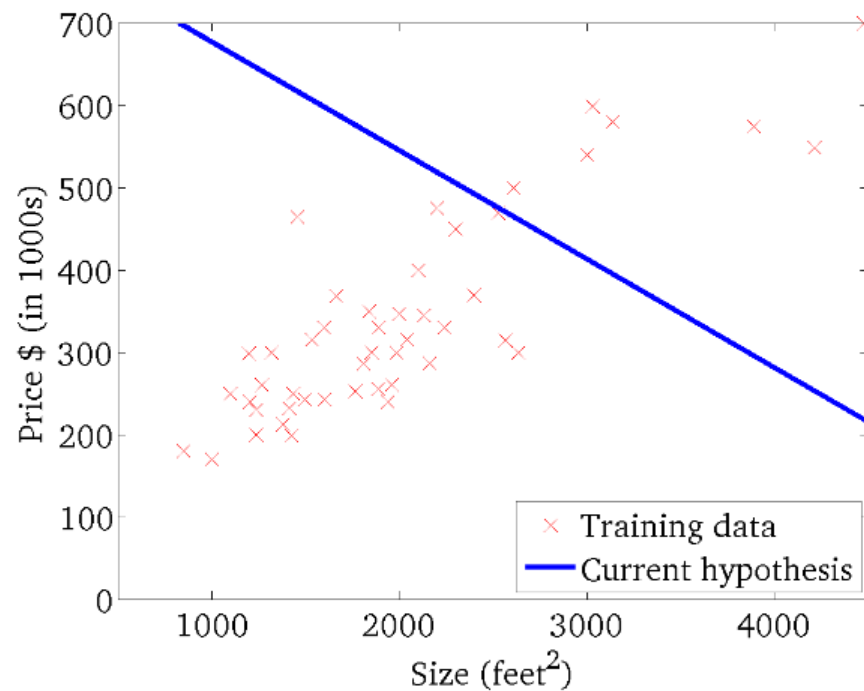


→ Surface plot.

Behaviour for different values of parameters

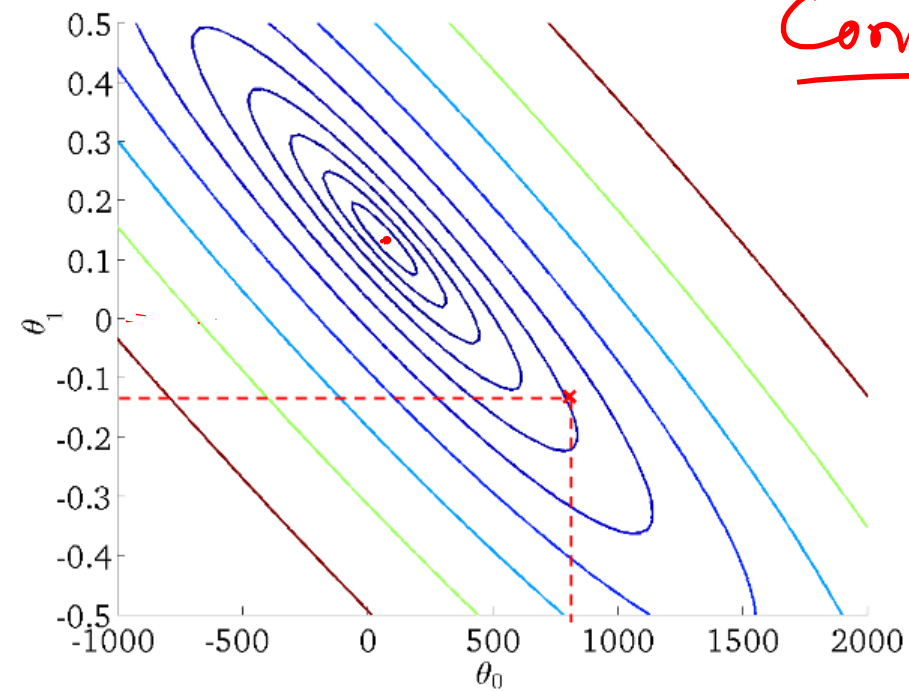
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



$$J(\theta_0, \theta_1)$$

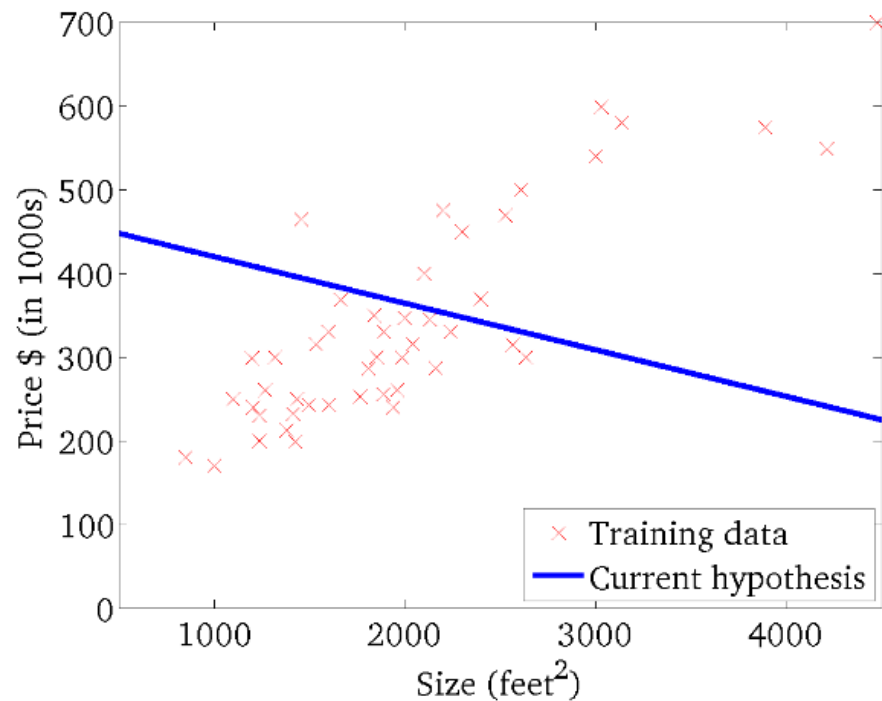
(function of the parameters θ_0, θ_1)



Behaviour for different values of parameters

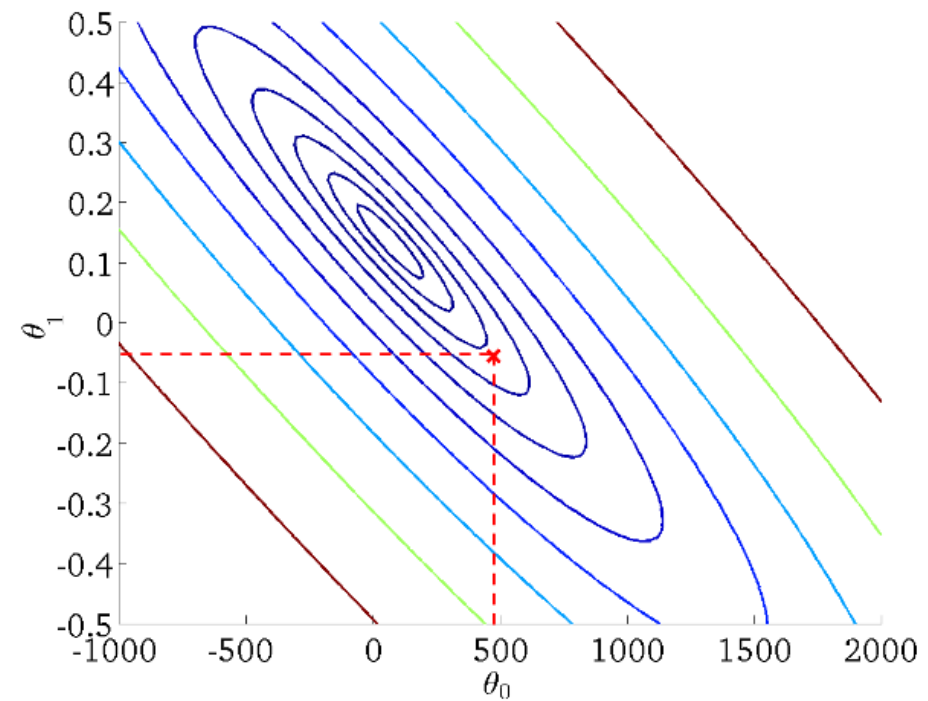
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)

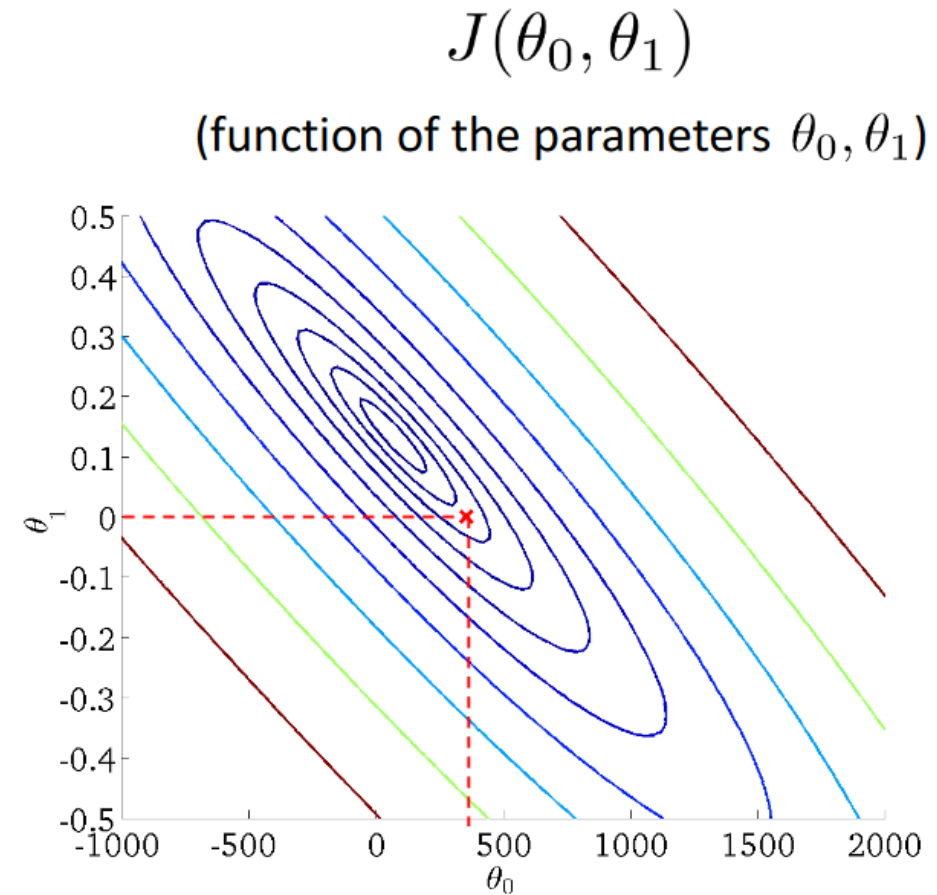
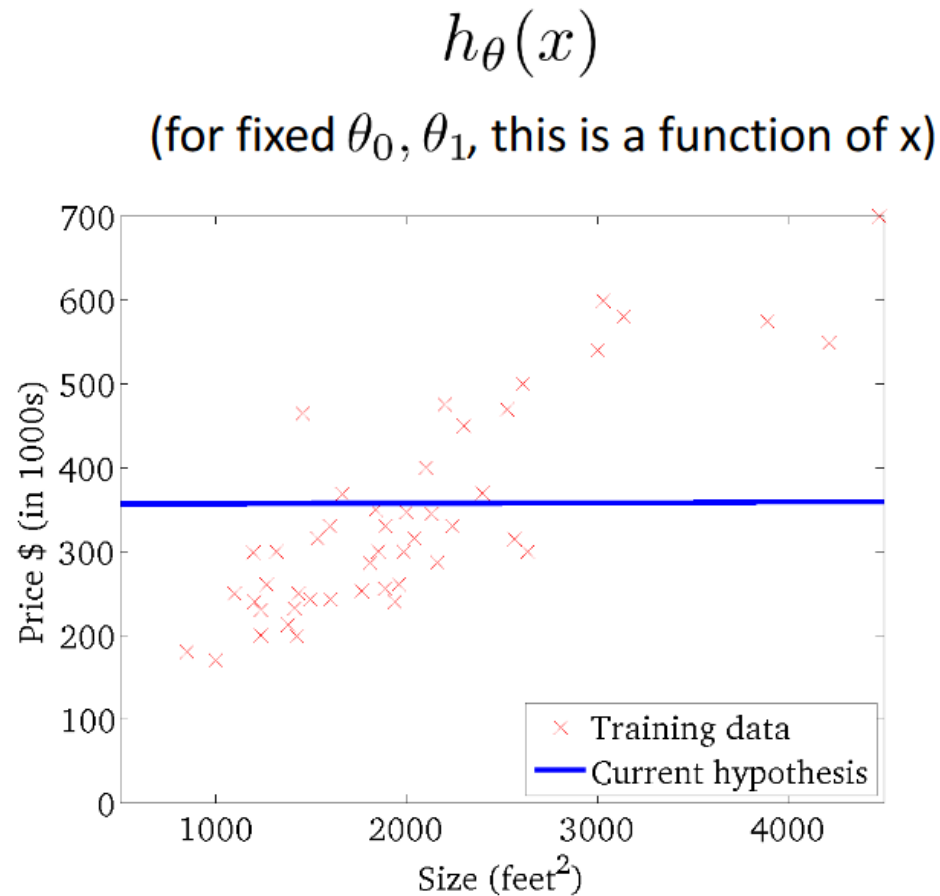


$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)



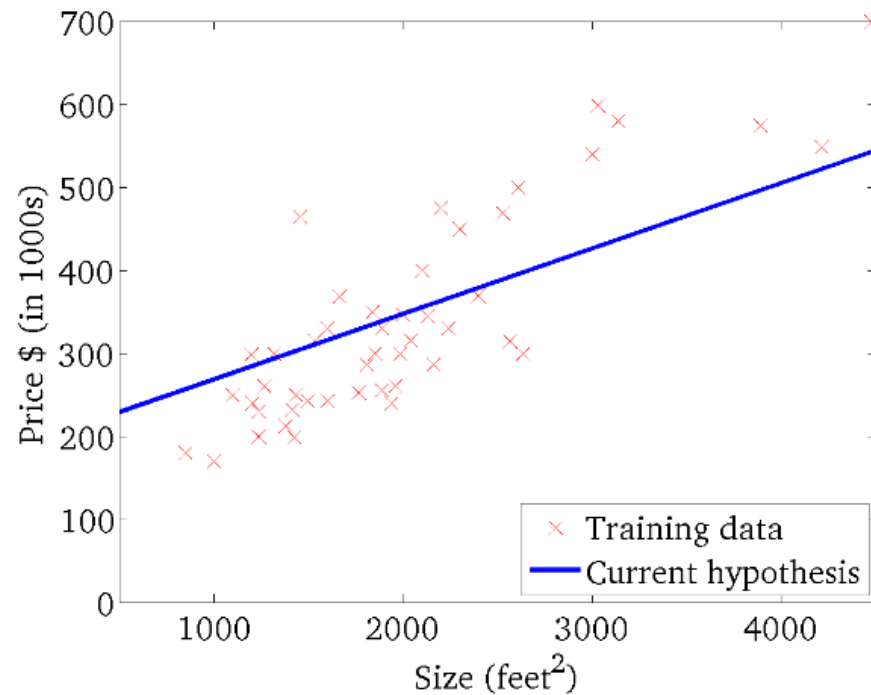
Behaviour for different values of parameters



Behaviour for different values of parameters

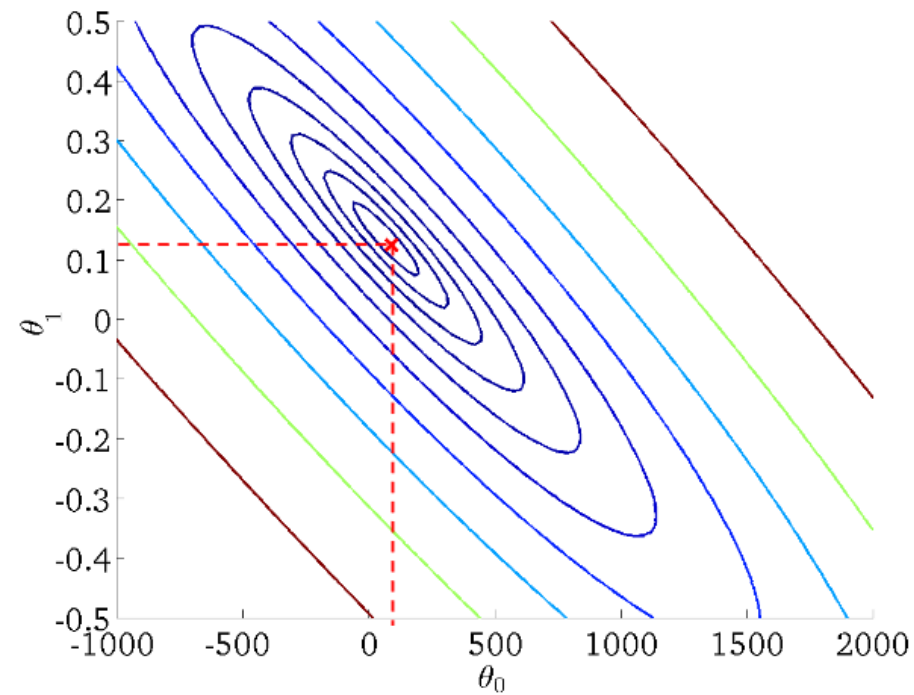
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



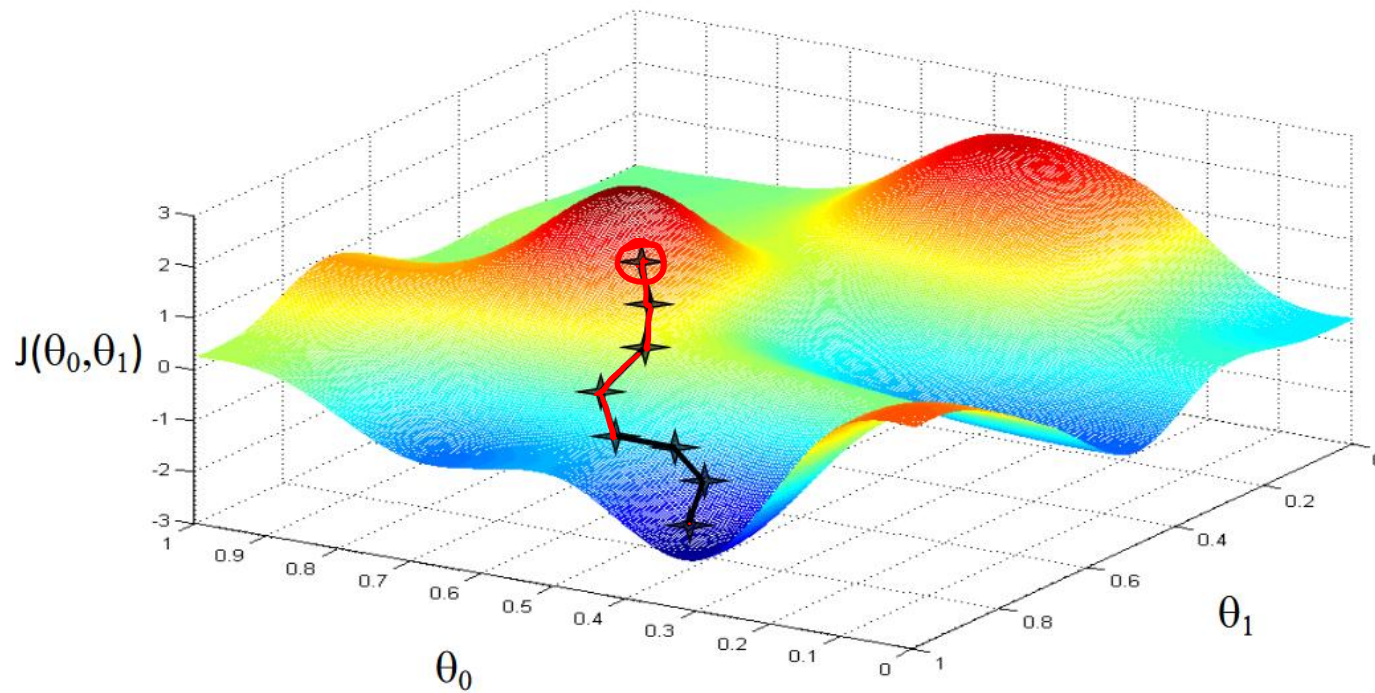
$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)



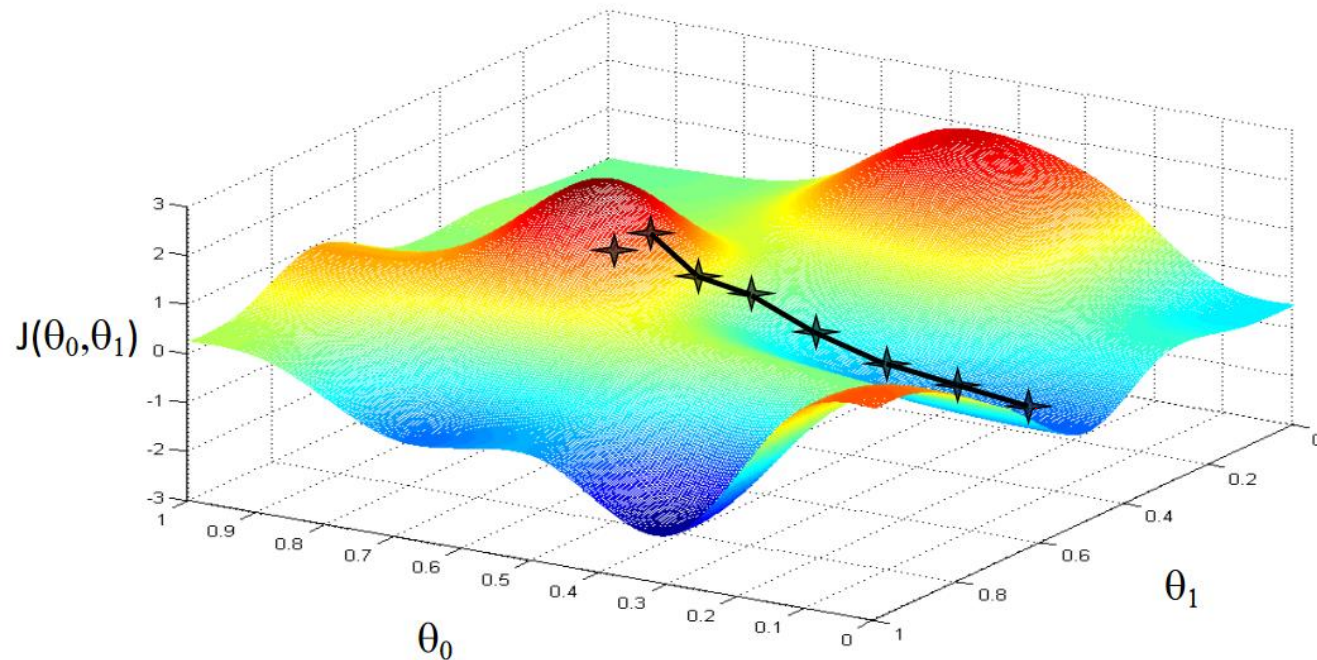
Reaching a minimum

- Choose initial value for θ
- Chose a new value for θ to reduce $J(\theta)$



Reaching a minimum

- Choose initial value for θ
- Chose a new value for θ to reduce $J(\theta)$



Gradient Descent

$x = 1$

d
 ∂

- Initialize θ ← d-dimensional vector
- Repeat until convergence

Learning Rate.

$$\theta_j \boxed{:=} \theta_j - \boxed{\alpha} \frac{\partial J(\theta)}{\partial \theta_j}$$

← subtr

compute

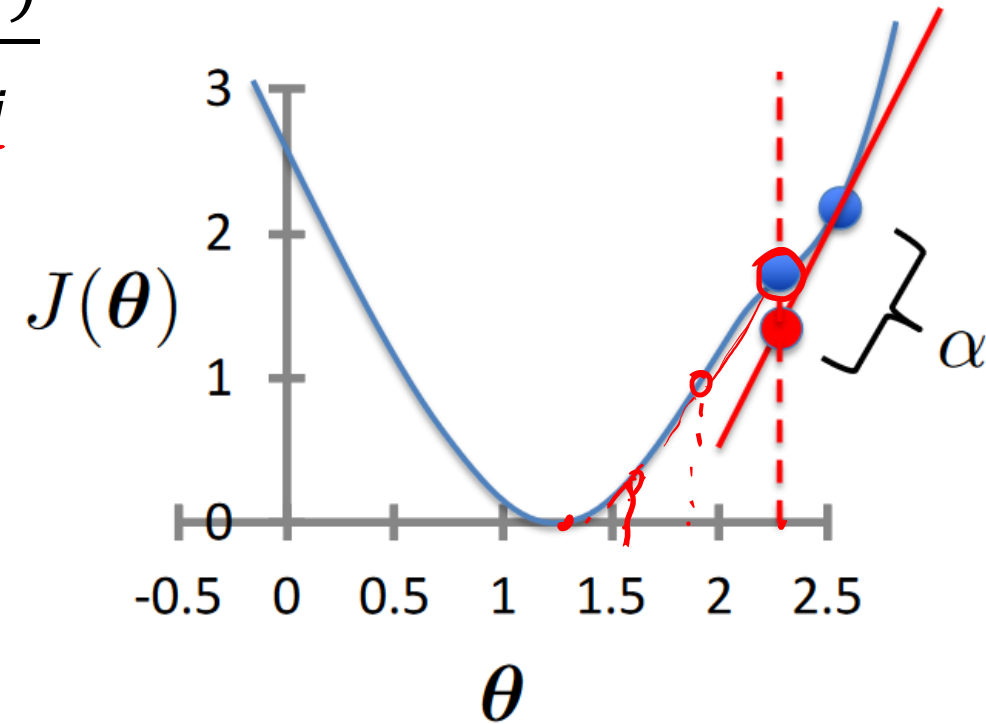
$$\theta_{j(k+1)} = \theta_{j,k} - D\theta$$

$$t_0 := \theta_0 - \alpha \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$$

$$t_1 := \theta_1 - \alpha \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1)$$

$$\theta_0 := t_0$$

$$\theta_1 := t_1$$



$$t_0 := \theta_0 - \alpha \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$$

$$\theta_0 := t_0$$

$$t_1 := \theta_1 - \alpha \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1)$$

$$\theta_1 := t_1$$

Incorrect way